

Effects of coffee pulp in the diets of growing lambs on growth, meat quality, and blood parameters

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Abstract

This study aimed to evaluate the effects of including varying amounts of coffee pulp (COP) in the diet on the growing lambs' meat quality, growth performance, nutrient digestibility, and slaughtering performance. In a random design, twenty-four Awassi lambs (body weight: 16.7 ± 1.36 kg) were divided into three treatment diets, each containing eight. Experimental diets contained 0 (CON), 50 (COP50), or 100 (COP100) g of COP per kg DM. The study lasted 80 days, with the first 10 days being used to acclimatize the lambs in the pens and diets, followed by 70 days for data collection. The results showed that intakes of dry matter (DM), crude protein (CP), and neutral detergent fiber (aNDF) were comparable among the three treatment groups. While the acid detergent fiber (ADF) consumption of lambs fed COP50 was similar to that of lambs fed CON and COP100, the ADF intake of lambs in the COP100 group was considerably greater ($P \leq 0.04$) than that of the CON. Lambs' initial, final, total, and average daily gain weights were similar across treatment diets. Adding the COP did not affect the feed efficiency. In contrast to the CON diet, the cost gain for the COP diets were significantly lower ($P < 0.05$). The digestibility of DM, CP, aNDF, and ADF, as well as N balance parameters did not differ among the three treatment diets ($P \geq 0.15$). Including of COP in the diet had no effect on the lambs' slaughter performance or meat quality metrics. Additionally, adding COP to the diet did not affect blood serum content such glucose, urea N, lipid profile, and liver enzymes. In summary, supplementing lambs' diets with 50 or 100 g/kg of COP showed similar results in feed intake, digestibility, growth, slaughter performance, meat quality, and health. Crucially, COP up to 100 g/kg DM reduced production costs and proved to be a viable alternative feed source.

Introduction

Given the escalating costs of conventional animal feeds and the adverse environmental impacts of agro-industrial by-products, adopting appropriate alternative feeds in animal feeding systems would be advantageous in maintaining the productivity and sustainability of livestock production systems and alleviating environmental pollution (Obeidat, 2025).

In this context, Coffee (*Coffea sp.*) is one of the most significant agricultural products in the world (Ma et al., 2022, Mendes dos Santos et al., 2021), and coffee harvesting and processing can lead to producing valuable by-products that can be exploited as ruminant feeds. The two main coffee varieties grown for commercial production worldwide are *Coffea arabica* and *Coffea robusta* (Mussatto et al., 2011, Pandey et al., 2000). These two varieties vary in sensory, physical, and chemical characteristics (Alves et al., 2009). Around 12.2 million hectares of coffee were grown globally in 2023, yielding 11.1 million tons (FAO Food and Agriculture Organization of the United Nations, 2023). In 2023, Brazil was the largest producer, with 3.41 million tons, followed by Vietnam (2 million tons), Indonesia (0.76 million tons), Colombia (0.68 million tons), and Ethiopia (0.56 million tons). Most of this production came from developing nations, especially those with tropical and subtropical climates (FAO Food and Agriculture Organization of the United Nations, 2023).

Coffee can be exposed to either wet or dry processing methods, resulting in different by-products (Mendes dos Santos et al., 2021, Oliveira et al., 2021, Klingel et al., 2020, Mazzafera, 2002, Pandey et al., 2000). Coffee by-products have emerged as one of the most significant possible sources of nutrients for animal feed, mainly because of their nutritional composition and cheaper cost than conventional feeding ingredients (Pedraza-Beltrán et al., 2012). The pulp is the first main by-product produced during the wet processing of coffee cherries (Oliveira et al., 2021). During processing, for every 2 tons of processed coffee, 1 ton of coffee pulp (COP) is obtained (Mendes dos Santos et al., 2021). The COP makes up between 40 and 50 % of the weight of the coffee cherry (Marín-Tello et al., 2020), representing 29 % of the dry weight of whole berry (Mendes dos Santos et al., 2021, Klingel et al., 2020). On a dry weight basis, COP comprised of high levels of carbohydrates (210–320 g/kg), proteins (50–150 g/kg), total pectic compounds (65 g/kg), minerals (90 g/kg), lipids (20–70 g/kg) (Blinov ´ a et al., 2017, Mendes dos Santos et al., 2021) that may favor its inclusion in animal diets.

A previous study by Maxiselly et al. (2022) concludes that including COP at a level of up to 200 g/d in the goat diet may improve the efficacy of animal digestion and rumen fermentation without affecting feed intake, rumen microbes, or blood metabolites. Hernández-Bautista et al. (2018) concluded that the productive parameters of sheep were not affected by including up to 280 g/kg COP in the diet. A study by Pedraza-Beltrán et al. (2012) observed that including the COP at a rate of 200 g/kg in the concentrate for cows did

not affect their milk yield, milk composition, and grass DM intake while reducing the cost of feeding. According to Noriega et al. (2008), adding 150 g/kg COP had no effect on sheep performance in terms of growth, and the same inclusion level of coffee pulp treated with urea and soybean seed did not affect carcass weight. A sheep study showed that adding up to 160 g/kg of COP containing caffeine and phenolic acids improved carcass dressing and reduced rumen and intestinal fat without changing the meat's oxidative stage (Salinas-Rios et al., 2014a). However, previous studies reported that COP contains anti-nutritional factors such as caffeine, polyphenols, and tannins that may limit animal utilization (Mendes dos Santos et al., 2021, Franca and Oliveira, 2009, Farah and Donangelo, 2006). These anti-nutritional factors differ based on the coffee variety, growth stage, cultivation process, and extraction method (Pedraza-Beltrán et al., 2012).

Livestock production in many tropical and subtropical regions uses large quantities of commercial concentrates. In Jordan, high feed costs of up to 75 % (Al-Khaza'leh, 2020) make small ruminant production an expensive enterprise, affecting the profitability of smallholder farmers. An accumulation of thousands of tons of underutilized COP each year can contribute to significant adverse environmental issues for coffee-producing countries (Shuai et al., 2023, Mendes dos Santos et al., 2021, El Achaby et al., 2019, Pedraza-Beltrán et al., 2012). Although there is growing interest worldwide in adding agro-industrial by-products (e.g. COP) to ruminant diets, there is still a dearth of empirical data regarding the safety and effectiveness of incorporating in small ruminant production systems, especially in middle east region where there are different animal breeds, production systems, and environmental factors. Furthermore, although the investigation of COP has been studied in a number of livestock species, there are few studies assessing its effects with different ratios on meat quality, carcass characteristics, and nutrient consumption in Awassi lambs, a common fat-tailed breed in Jordan. The study hypothesizes that including up to 50 and 100 g/kg COP in the diet would improve feed intake and digestibility and, consequently, growth performance, carcass characteristics, and meat quality of growing Awassi lambs. Therefore, Awassi lambs were assessed for the effects of COP inclusion on nutrient intake, digestibility, and N balance, blood parameters, BW changes, carcass characteristics, and meat quality.

Animals, housing, and experimentation

Twenty-four lambs with an initial mean body weight of 16.7 ± 1.36 kg were assigned randomly to one of three diets. Lambs were purchased from a local rancher and then shipped to an Animal Farm facility within the Jordan University for Science and Technology (JUST). The climatic conditions of the study are semi-arid, located at a latitude of 32°30'N

and an elevation of 510 m above sea level. Before the study began, JUST's Animal Care and Use Committee reviewed and approved all protocols. Lambs

Results

Including COP did not affect the nutrient content of diets (Table 1). The cost of feed for the CON diet, however, was the highest (433 US\$/ton), followed by COP50 (391 US\$/ton) and COP100 diet (371 US\$/ton).

The impact of COP on nutrient intake is shown in Table 2. Intake of DM, CP, and aNDF was similar for the three treatment groups. However, lambs in the COP100 group consumed significantly more ($P = 0.04$) ADF than lambs fed CON, while lambs fed COP50 consumed ADF like lambs fed CON and COP100.

Discussion

In the present research, there were no variations in the chemical content of the experimental diets. The chemical content of COP has been reported to vary based on the coffee varieties (Marín-Tello et al., 2020) and treatment processes (Salinas-Rios et al., 2014b). Our study depicts a strong correlation between the cost of feeds and the quantity of COP added to Awassi lamb diets. The findings corroborate earlier studies that utilized substitute feeds (e.g., COP) to lower the cost of traditional

Conclusion

In summary, adding up to 100 g/kg of coffee pulp (COP) in the diet of growing Awassi lambs reduced production costs and provided a promising alternative feed source without negatively affect nutrient digestibility, growth and slaughter performance, meat quality, or overall health.

CRedit authorship contribution statement

Hadil S. Subih: Writing – review & editing, Writing – original draft, Formal analysis. **Obeidat Belal:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jafar Al-Khazaleh:** Writing – review & editing, Writing – original draft, Formal analysis.

Declaration of Competing Interest

There is no conflict of interest with anyone regarding this manuscript.

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Credit author statement

Belal S. Obeidat, Jafar Al-Khazaleh, and Hadil S. Subih designed, conducted, writing the first and final manuscript.